

Cheap Multistatic Radar: Passive Illumination, Adaptive Cancellation, and Low-Cost Coherent Nodes

Bjørn Remseth
Email: la3lma@gmail.com
With AI assistance

Abstract—This concept document explores a cheap multistatic radar architecture that uses existing transmitters, such as digital television, cellular base stations, and other radio-frequency illuminators of opportunity, together with low-cost coherent receiver nodes. The central idea is to dedicate one receiver channel to a strong reference copy of the illuminator signal, use surveillance channels to observe the scene, and run streaming adaptive cancellation near the radio front end so the direct-path and static multipath energy are suppressed before weaker moving echoes are searched in delay-Doppler space. The document outlines the physics, bistatic geometry, cross-ambiguity processing, adaptive-filter mathematics, analog-to-digital dynamic-range constraints, FPGA/GPU compute split, and a transparent cost model for prototype and production paths. It is a concept and engineering sketch, not a procurement recommendation or validated product design.

Index Terms—passive radar, multistatic radar, passive coherent location, adaptive filtering, direct-path interference cancellation, software-defined radio, FPGA, drone detection

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Note on document creation: This document was created with AI assistance. See the “About This Document” section for details on the methodology and verification processes used.

I. EXECUTIVE TL;DR

Concept: Build a distributed passive/multistatic radar network from inexpensive coherent receiver nodes. Each node listens to a strong broadcast or cellular signal in a reference channel, listens to the sky or protected area in surveillance channels, adaptively subtracts the direct-path and stationary clutter energy in streaming hardware, and reports compact delay-Doppler observations rather than raw radio data.

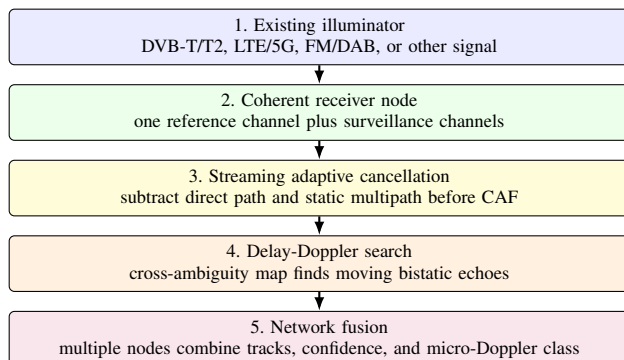


TABLE I
First-order conclusions for a cheap multistatic radar concept.

Issue	Design implication
Illuminator	Use existing signals first; controlled transmitters create regulatory and detectability burdens.
Bandwidth	Range-delay resolution scales as c/B . LTE-like 20 MHz signals are much better than narrowband signals.
Dynamic range	8-bit radios are a serious bottleneck near strong transmitters. A 12–14 bit coherent receiver is a more plausible prototype baseline.
Cancellation	Direct-path interference should be cancelled before the expensive delay-Doppler search. Streaming FPGA logic is a natural fit.
GPU	A large GPU helps with the CAF, tracking, and classification, but cannot recover echoes lost to analog clipping or quantization.
Detection	Birds and drones may overlap in range and Doppler; micro-Doppler and track behavior are needed for classification.

The most credible prototype path is not a finished custom board. It is a staged experiment: start with coherent software-defined radios, prove cancellation and delay-Doppler processing against controlled targets, then move to a dedicated board once the signal-processing and dynamic-range assumptions are measured.

II. PROBLEM DEFINITION

The system should detect and track small moving objects, including drones, without relying on a centralized high-power radar transmitter. Instead, it exploits radio signals that already illuminate the scene. Examples include terrestrial digital television, cellular base stations, FM or DAB broadcast, and possibly site-owned communication transmitters. This makes the radar node passive in the RF-emission sense when it uses only illuminators of opportunity, although the whole system is still an active sensing system in the broader operational sense.

The hard problem is that the target echo is tiny compared with the direct-path signal from the transmitter and compared with static multipath from buildings, terrain, towers, and nearby structures. A receiver may see a broadcast tower or cellular base station as an enormous signal while the drone echo is tens of decibels weaker. The engineering challenge is therefore not only radar processing. It is preserving the weak

echo through the antenna, analog front end, ADC, clocking, cancellation, and compute pipeline.

III. ENGINEERING REALITY CHECK

The starting engineering intuition for this report is blunt: a cheap passive radar node is not made real by attaching a large GPU to a cheap radio. A GPU can accelerate delay-Doppler search, image formation, clustering, tracking, and classification. It cannot repair analog clipping, missing ADC codes, badly chosen gain, clock phase noise, reference-channel contamination, poor antenna placement, or geometry errors.

The practical constraints divide into two classes:

- **Digital compute constraints:** CAF size, coherent processing interval, number of delay and Doppler bins, number of receiver channels, classifier cost, and network fusion.
- **Physics and front-end constraints:** direct-path interference, multipath, ADC dynamic range, RF filtering, gain distribution, oscillator stability, antenna isolation, and calibration.

The “throw a 5090 at it” hypothesis is therefore only half true. High-end GPUs can make the expensive radar math less frightening. They do not trivialize the receiver. If the surveillance channel is already clipped by the illuminator leakage, or if the target echo falls below the quantization floor, the later CAF has no valid target information to recover.

The engineering response is to push mitigation earlier in the chain:

- 1) use physical antenna placement, directivity, terrain shielding, and RF filtering to reduce direct-path energy before the LNA and ADC;
- 2) choose enough ADC resolution and front-end linearity that weak echoes survive digitization;
- 3) run adaptive cancellation close to the ADC stream, ideally in FPGA logic, so the direct path and static multipath are reduced before bulk data movement and CAF processing;
- 4) reserve GPU compute for the high-dimensional search, micro-Doppler features, and fusion work where it genuinely helps.

Explainer: passive coherent location. Passive coherent location, or PCL, is radar processing that uses a non-cooperative transmitter as the illuminator. The receiver keeps a reference copy of the transmitted waveform and correlates it with surveillance-channel echoes. A moving target appears as a delayed and Doppler-shifted copy of the reference signal. The receiver did not transmit the waveform, but it must still solve the ordinary radar problems of geometry, clutter, timing, dynamic range, and false alarms.

IV. BISTATIC AND MULTISTATIC GEOMETRY

In monostatic radar, the transmitter and receiver are collocated. In bistatic radar, the transmitter and receiver are separated. In multistatic radar, there are multiple transmitters,

receivers, or both. The basic passive-radar measurement is an excess path length:

$$\rho(\mathbf{x}) = \|\mathbf{x} - \mathbf{t}\| + \|\mathbf{x} - \mathbf{r}\| - \|\mathbf{t} - \mathbf{r}\|, \quad (1)$$

where $\mathbf{t}$ is transmitter position, $\mathbf{r}$ is receiver position, and $\mathbf{x}$ is target position. The corresponding delay is

$$\tau(\mathbf{x}) = \frac{\rho(\mathbf{x})}{c}, \quad (2)$$

where c is the speed of light. A single delay measurement constrains the target to an ellipsoid in three dimensions, or an ellipse in a two-dimensional slice. Multiple receivers, multiple transmitters, angle estimates, and track continuity reduce the ambiguity.

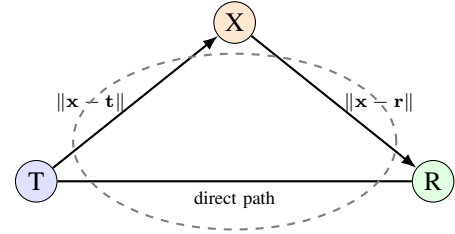


Fig. 1. Bistatic geometry. The target echo follows transmitter-target-receiver path length, while the reference channel sees the direct transmitter-receiver path.

The bistatic Doppler frequency is driven by the time derivative of the total transmitter-target-receiver path length. A useful approximation is

$$f_D \approx -\frac{1}{\lambda} \mathbf{v} \cdot (\hat{\mathbf{u}}_t + \hat{\mathbf{u}}_r), \quad (3)$$

where λ is wavelength, $\mathbf{v}$ is target velocity, $\hat{\mathbf{u}}_t$ points from target toward transmitter, and $\hat{\mathbf{u}}_r$ points from target toward receiver. The sign convention varies by implementation; the physical point is that bistatic Doppler depends on both legs of the path, not only radial velocity relative to the receiver.

V. ILLUMINATORS AND RESOLUTION

For a waveform with effective bandwidth B , the delay resolution is approximately

$$\Delta\rho \approx \frac{c}{B}. \quad (4)$$

Radar texts often quote monostatic range resolution $c/(2B)$. Passive bistatic processing more naturally measures excess path length, so both values should be kept explicit. Table II gives first-order values.

This explains why DVB-T/T2 and LTE-like signals are attractive. DVB-T and DVB-T2 are OFDM systems standardized by ETSI [1], [2]. LTE physical channels are standardized by 3GPP [3]. Their structure is not designed for radar, but wide, structured, high-duty-cycle signals can be exploited by correlation processing.

TABLE II

Bandwidth-driven delay resolution. Values are idealized; real performance also depends on waveform ambiguity, SNR, clutter, integration time, and geometry.

Signal bandwidth	c/B	c/(2B)
200 kHz	1500 m	750 m
1.5 MHz	200 m	100 m
8 MHz	37.5 m	18.75 m
20 MHz	15 m	7.5 m

VI. SIGNAL MODEL

Let $x[n]$ be the sampled reference-channel copy of the illuminator. A surveillance channel can be modeled as

$$y[n] = \sum_{k=0}^{K-1} a_k x[n-d_k] + \sum_{m=1}^M \alpha_m x[n-d_m] e^{j2\pi f_m n T_s} + w[n]. \quad (5)$$

The first sum represents direct-path leakage and static multipath. These terms may be huge but mostly stationary over short intervals. The second sum represents moving target echoes with delay d_m , Doppler f_m , and complex amplitude α_m . The noise term is $w[n]$.

The classic passive-radar detection product is the cross-ambiguity function:

$$\chi(\tau, \nu) = \sum_{n=0}^{N-1} y[n] x^*[n-\tau] e^{-j2\pi \nu n T_s}. \quad (6)$$

Targets appear as peaks in delay-Doppler space. Unfortunately, direct-path and static multipath energy can dominate the map. A cheap system therefore needs cancellation before or during the CAF.

VII. ADAPTIVE CANCELLATION

The adaptive canceller builds a local model of the interference in each surveillance channel from delayed copies of the reference channel:

$$\hat{y}_{\text{dpi}}[n] = \mathbf{w}^H[n] \mathbf{x}_n, \quad (7)$$

where

$$\mathbf{x}_n = [x[n] \quad x[n-1] \quad \cdots \quad x[n-L+1]]^T. \quad (8)$$

The residual is

$$e[n] = y[n] - \mathbf{w}^H[n] \mathbf{x}_n. \quad (9)$$

The least-mean-squares update is

$$\mathbf{w}[n+1] = \mathbf{w}[n] + \mu \mathbf{x}_n e^*[n], \quad (10)$$

with step size μ . More advanced algorithms, such as normalized LMS or recursive least squares, may converge faster or behave better under changing channels, but LMS is the right mental model for a first streaming FPGA design. Adaptive filtering is a standard signal-processing technique [4], and

direct-path interference cancellation is a central problem in passive radar [5], [6].

VIII. WHY ADC BITS MATTER

A large GPU cannot recover a target echo that was clipped or quantized away before digitization. The ideal quantization signal-to-noise ratio of an N -bit ADC is approximately

$$\text{SNR}_q \approx 6.02N + 1.76 \text{ dB}. \quad (11)$$

TABLE III

Ideal ADC quantization dynamic range. Real receivers are worse after analog imperfections, gain settings, clock jitter, and front-end overload.

ADC bits	Ideal quantization SNR
8	49.9 dB
10	62.0 dB
12	74.0 dB
14	86.0 dB
16	98.1 dB

This is the first harsh engineering constraint. Low-cost coherent receivers such as KrakenSDR are attractive because they provide multiple coherent channels at low cost, but their RTL-SDR-class lineage is not a high-dynamic-range radar front end [7]. They may still be useful for learning, geometry tests, and friendly signal conditions. For a serious passive radar prototype near strong illuminators, 12–14 bit coherent receivers are much more plausible.

The ADALM-Pluto uses an AD9363 RF transceiver and Zynq-7000 processing system [8], [9]. It is not a finished passive radar node, but it is a useful reference class: a low-cost SDR with a more capable mixed-signal front end and FPGA fabric in the path. Multiple Pluto-class receivers locked to a shared reference can form a practical lab prototype. Higher-end Ettus/USRP-class hardware can be used when synchronization, bandwidth, or dynamic range dominate the experiment.

IX. COMPUTE BUDGET

Consider four coherent complex channels sampled at 20 MS/s. If each complex sample is represented as 16-bit I and 16-bit Q, the raw stream is

$$4 \times 20 \times 10^6 \times 4 \text{ bytes/s} = 320 \text{ MB/s}. \quad (12)$$

That is before metadata, buffering, disk writes, or network overhead. A node that forwards all raw samples is therefore wasteful. It should reduce the stream locally.

For a 64-tap complex LMS canceller on three surveillance channels at 20 MS/s, the approximate complex multiply-accumulate rate is

$$3 \times 64 \times 20 \times 10^6 = 3.84 \times 10^9 \quad (13)$$

complex MACs per second, plus coefficient updates. This is a streaming workload with fixed dataflow and tight latency.

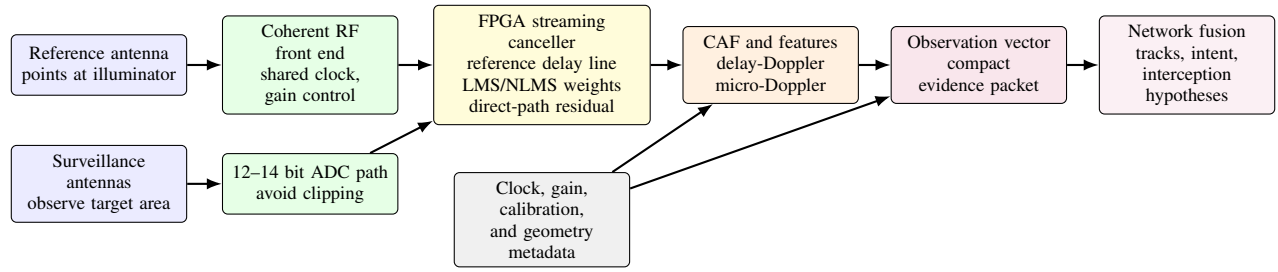


Fig. 2. *Candidate processing chain. The node reduces raw RF streams into compact observation vectors. Network fusion is shown as the downstream consumer: it combines weak, sporadic, multi-node evidence into tracks, intent hypotheses, and possible interception opportunities.*

It belongs naturally in FPGA logic, not in a general-purpose CPU loop.

The CAF can then operate on residuals, selected bands, decimated streams, or coherent processing intervals. GPUs are useful here because delay-Doppler search, thresholding, clustering, micro-Doppler spectrograms, and neural classifiers are parallel workloads. The correct division is therefore:

- **FPGA:** high-rate stream alignment, delay lines, adaptive cancellation, decimation, packetization.
- **GPU/CPU:** CAF batches, tracking, classifier inference, networking, storage, UI, and fleet management.

Explainer: observation vectors and network fusion. An observation vector is a compact evidence packet: the useful information extracted by one node during one processing interval, expressed in a form that can be refined further up the processing chain. It is a generic systems term, not a radar standard. In this report it means something like “a weak moving thing may be present here, with this bistatic delay, Doppler, confidence, micro-Doppler evidence, clock quality, and calibration context.”

Network fusion is the consumer of those vectors. It correlates sparse observations from multiple nodes and illuminators, maintains target tracks, rejects inconsistent evidence, estimates confidence, and can run AI-assisted intent analysis or interception planning. From the sensor-node design point of view, fusion is not the thing being built first. It is the demanding downstream customer that defines what evidence the node must preserve and report.

X. BIRDS, DRONES, AND MICRO-DOPPLER

Small drones and birds may have overlapping radar cross section, speed, and altitude. A detector that only thresholds moving range-Doppler peaks will produce false alarms. Classification requires additional features.

Micro-Doppler is the time-varying Doppler structure caused by rotating propellers, flapping wings, vibration, and body motion. Drone rotors can produce high-rate spectral sidebands or comb-like energy around the body Doppler; bird wings tend to produce lower-rate periodic structure. Micro-Doppler classification of drones is an active research area [10].

The implication for this concept is that the node should not report only a point detection. It should report an observation vector:

- transmitter and receiver identities,
- bistatic delay and Doppler,
- SNR and cancellation residual metrics,
- bearing if an antenna array is available,
- micro-Doppler features or evidence snippets,
- clock and calibration quality,
- local environment and node health metadata.

XI. CANDIDATE HARDWARE ARCHITECTURE

The first experimental architecture should be modular:

- **Reference antenna:** directional or high-SNR path toward the illuminator.
- **Surveillance antennas:** three or more antennas observing the protected area.
- **Coherent receiver:** shared clock and phase-coherent channels.
- **Timing:** GPSDO or other disciplined reference for multi-node experiments; u-blox timing modules such as the ZED-F9T class are relevant candidates [11].
- **FPGA:** streaming cancellation, decimation, and packetization.
- **Edge compute:** Jetson, x86, or similar compute for CAF, classification, storage, and networking.
- **Backhaul:** Ethernet, LTE, or other encrypted link to a fusion server.

The economic claim should be modest: custom integration can reduce unit cost dramatically after the signal-processing concept is proven. The exact COGS is not knowable from architecture alone. It depends on bandwidth, channel count, environmental rating, clock requirements, manufacturing volume, radio bands, compute placement, and test yield.

TABLE IV

Prototype and production cost model. These are planning assumptions, not vendor quotes. Numbers should be replaced by sourced quotes before procurement decisions.

Subsystem	Prototype assumption	Volume assumption	Notes
Coherent RF receiver	\$1000–\$4000	\$80–\$160	Prototype may use multiple Pluto-class SDRs or higher-end lab SDRs. Volume node would integrate RF transceivers.
FPGA/SoC compute	\$150–\$800	\$30–\$80	Prototype can use SDR-integrated FPGA plus edge computer. Volume depends strongly on logic size and memory.
Clock/timing	\$150–\$1000	\$20–\$60	GPSDO, disciplined oscillator, distribution, and holdover requirements drive cost.
Antennas and RF front end	\$100–\$800	\$30–\$100	Includes filters, LNAs, limiters, splitters, cables, and antenna mechanics.
Edge compute and storage	\$100–\$800	\$20–\$100	May disappear into SoC or remain as module depending on CAF/classifier placement.
Power, enclosure, assembly	\$150–\$1000	\$50–\$150	Weatherproofing, thermal path, connectors, and mounting dominate field units.
Total per node	\$1650–\$8400	\$230–\$650	High uncertainty until requirements and volumes are real.

XII. PROTOTYPE PATH

There are two valid experimental strategies. They answer different questions and should be run in parallel if budget allows.

A. Going From The Top

The top-down strategy starts with hardware that is too good and too expensive for the final system: high-dynamic-range SDRs, disciplined clocks, carefully characterized antennas, low-loss cables, and enough GPU/CPU capacity that compute is not the first bottleneck. Ettus USRP-class devices and other lab SDRs are useful in this lane because they reduce uncertainty around bandwidth, ADC resolution, clocking, and software tooling [12], [13]. BladeRF-class devices are another intermediate option with accessible 2x2 MIMO SDR hardware and FPGA fabric [14].

The purpose is to prove the physics and algorithms:

- Which illuminators produce usable ambiguity functions?
- How much cancellation is required before targets become visible?
- What coherent integration time is practical before phase drift or target motion smears the peak?
- Which micro-Doppler features distinguish drones, birds, and clutter?
- What observation vector should a node report?

This path is allowed to be expensive because it avoids premature optimization. It gives the project an upper-bound reference: if the concept fails even with good radios and clocks, a cheap product will not save it.

B. Going From The Bottom

The bottom-up strategy starts with the cheapest coherent hardware and discovers exactly where it breaks. KrakenSDR-class hardware is useful here because it provides five coherent-capable channels at low cost and is explicitly marketed for direction finding and passive-radar experimentation [7], [15]. Its dynamic-range and bandwidth limitations should be treated as experimental variables, not embarrassments.

The purpose is to discover the minimum viable node:

- Can antenna geometry and passive shielding reduce the direct path enough for an 8-bit coherent receiver?
- How much can analog cancellation buy before the ADC?
- Which illuminators and geometries are forgiving enough for cheap hardware?
- Can FPGA or near-edge cancellation make a low-cost receiver useful after modest analog protection?
- What is the cheapest receiver that still preserves enough echo information?

The bottom-up path is valuable because the eventual system goal is cheap and distributed. It prevents the design from becoming a beautiful lab instrument that cannot scale. But it must be honest: if measurements show that 8-bit receivers erase the target in realistic geometries, the design should move to 12-bit or 14-bit hardware rather than wish the problem away.

C. Converging The Two Paths

The two approaches should meet in the middle, with simulation acting as the cheapest shared testbench between them. The top-down lane establishes what good looks like. The bottom-up lane establishes what cheap can survive. The production design should inherit the algorithms, calibration, and observation schema from the top-down lane, the cost discipline, ruggedization, and deployment lessons from the bottom-up lane, and the falsifiable parameter sweeps from simulation.

D. Phase 0: Offline Simulation

Build synthetic reference and surveillance channels with configurable direct-path interference, multipath, target delay, Doppler, oscillator error, ADC clipping, quantization, antenna isolation, target micro-Doppler, and clutter dynamics. This is not a disposable warm-up stage. It is the cheapest place to test principles, explore impossible-to-field combinations, and decide which hypotheses deserve laboratory time.

The simulation tooling must be treated as a first-class engineering artifact. A false positive can make the project chase a detector that only works because the simulator is too clean, too stationary, or too conveniently synchronized. A false

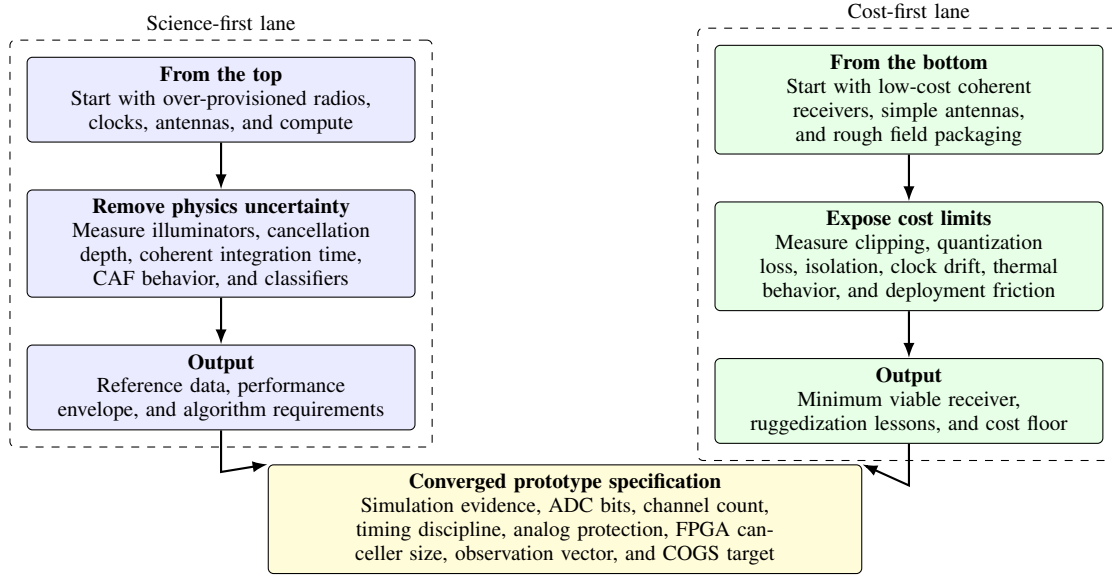


Fig. 3. The two prototype methodologies answer complementary questions. The top-down lane asks what performance is physically achievable when hardware is not the bottleneck. The bottom-up lane asks how cheap the node can become before the signal disappears or field deployment fails. The product architecture should be specified where the measured evidence from both lanes overlaps.

negative can kill a viable architecture because the simulator models the illuminator, propagation, receiver, or target motion pessimistically or incorrectly. Phase 0 should therefore include repeatable notebooks or scripts, saved parameter sets, controlled random seeds, unit tests for signal models, and deliberate adversarial cases. Its output is not merely a pretty range-Doppler plot; it is a ranked list of measurements that the top-down and bottom-up prototypes must validate.

Use this stage to choose cancellation length, ADC resolution, integration time, minimum target SNR, clock stability, and the first observation vector. Every strong claim from simulation should be tagged as either a design requirement, a lab-test question, or a field-test question.

1) *What Must Be Simulated:* The minimum useful simulator is a complex-baseband signal generator with enough realism to break naive algorithms. It should model the following layers before any field hardware is trusted:

- 1) **Illuminators:** LTE, DVB-T/T2, FM, DAB, Wi-Fi, or measured IQ files, including bandwidth, modulation, burst structure, spectral masks, and transmitter instability [3], [1], [2].
- 2) **Geometry and motion:** transmitter, receiver, target, and clutter positions over time, with bistatic range, bistatic Doppler, line-of-sight blockage, and target acceleration.
- 3) **Propagation:** direct-path leakage, static multipath, moving multipath, ground reflection, terrain or building masking, rain/fog attenuation where relevant, antenna gain patterns, and polarization mismatch.
- 4) **Receiver front end:** gain setting, noise figure, filter bandwidth, ADC resolution, clipping, phase noise, clock drift, sampling-rate offset, IQ imbalance, DC offset, and inter-channel skew.

- 5) **Processing chain:** adaptive cancellation, CAF or range-Doppler processing, CFAR-like detection, micro-Doppler extraction, observation-vector formation, and network-fusion handoff.

The first simulator can be deliberately modest: single illuminator, one reference channel, one surveillance channel, point targets, and a few static multipath taps. But it should already be good enough to act as a pre-antenna gate. Before the project connects receivers to real antennas in uncontrolled RF environments, the simulator should reproduce the expected scale of direct-path dominance, multipath structure, ADC headroom, cancellation depth, integration-time limits, and false-alarm mechanisms well enough that the first lab measurements have explicit predictions to confirm or falsify.

The simulator should also be designed as a progressively improving engineering asset, not a one-time feasibility script. Each experiment should feed it: measured illuminator IQ replaces idealized waveforms; measured receiver noise, clipping, clock drift, and IQ imbalance replace guessed impairment constants; antenna measurements replace isotropic gain assumptions; site surveys and field recordings replace generic multipath taps; and observed false alarms become regression cases. The model should therefore keep parameter files, raw measurement links, calibration provenance, and versioned scenario definitions. A better simulator is one that becomes less flattering as the project learns more, while still preserving simple canonical cases for debugging.

The high-fidelity simulator should eventually add realistic site geometry, antenna models, weather and clutter cases, multiple receivers, and target classes.

2) *Guiding Equations*: For receiver channel j , a useful starting model is

$$z_j[n] = \sum_p a_{jp} s[n - d_{jp}] e^{j\phi_{jp}[n]} + \sum_m \alpha_{jm}[n] s[n - d_{jm}[n]] e^{j2\pi\nu_{jm}nT_s} + c_j[n] + w_j[n], \quad (14)$$

where $s[n]$ is the illuminator waveform, the first sum represents direct path and static multipath, the second sum represents moving target echoes, d is delay in samples, ν is Doppler shift, $c_j[n]$ is structured clutter and interference, and $w_j[n]$ is receiver noise.

The simulated receiver should then apply practical damage:

$$r_j[n] = Q_B(\text{clip}(g_j z_j[n] + o_j[n])), \quad (15)$$

where g_j is gain, $o_j[n]$ captures receiver offset and leakage terms, $\text{clip}(\cdot)$ models analog or ADC saturation, and $Q_B(\cdot)$ models B -bit quantization. This is the equation that turns a beautiful radar idea into an engineering problem: if the direct path consumes the ADC range, the target can disappear before the adaptive filter ever sees it.

After adaptive cancellation produces a residual surveillance stream $e_j[n]$, the canonical passive-radar detection surface is a cross-ambiguity function:

$$\chi_j(\tau, \nu) = \sum_{n=0}^{N-1} e_j[n] x^*[n - \tau] e^{-j2\pi\nu nT_s}, \quad (16)$$

where $x[n]$ is the reference channel. Peaks in $\chi_j(\tau, \nu)$ are candidate observations, not final truth. A node-level observation vector can therefore be represented as

$$\mathbf{o}_{j,k} = [t_k, \hat{\tau}, \hat{\nu}, \widehat{\text{SNR}}, \eta_{\text{cancel}}, \mathbf{m}_{\text{micro}}, \mathbf{q}_{\text{clock}}, \Sigma], \quad (17)$$

where η_{cancel} summarizes cancellation quality, $\mathbf{m}_{\text{micro}}$ summarizes micro-Doppler or other classification features, $\mathbf{q}_{\text{clock}}$ summarizes timing quality, and Σ is an uncertainty estimate for fusion.

3) *Practical Tooling*: The practical tooling should be layered rather than monolithic:

- **Open reference model**: Python with NumPy and SciPy should be the first implementation for baseband modeling, FFTs, convolution, correlation, resampling, filters, Monte Carlo sweeps, and reproducible plots [16], [17]. This model is the audit trail.
- **Streaming DSP prototype**: GNU Radio should be used when the simulation needs to look like live SDR flowgraphs, replay captured IQ, or move toward hardware-in-the-loop without losing too much algorithm visibility [18].
- **GPU acceleration**: CuPy can accelerate large parameter sweeps, CAF grids, and Monte Carlo runs while keeping a NumPy-like programming model [19].
- **Radar-specialized commercial model**: MathWorks Radar Toolbox and Phased Array System Toolbox are

useful for cross-checking ambiguity processing, target signatures, phased-array assumptions, CFAR-style detection, and micro-Doppler behavior against mature radar tooling [20], [21].

- **Hardware-in-the-loop**: Analog Devices pyadi-iio is a practical bridge from Python simulations into Pluto/AD936x-class receivers, recorded IQ files, and repeatable receiver impairment measurements [22], [8], [9].
- **FPGA exploration**: AMD Vitis HLS is appropriate when the adaptive canceller, CIC/FIR stages, or streaming CAF pre-processing must be shaped into a candidate FPGA data path [23].
- **RF, antenna, and site fidelity**: Keysight RF System Design can model RF impairments and system-level radar chains, Ansys HFSS can model antennas and high-frequency structures, and Remcom Wireless InSite can support site-specific propagation and ray-tracing experiments [24], [25], [26].

The recommended practice is to keep the Python/NumPy/SciPy model as the source of conceptual truth, then use commercial or hardware-specific tools to falsify, calibrate, or refine it. When two tools disagree, the disagreement becomes an experiment: measure the real receiver, replay the same IQ through both chains, and update the model only when the cause is understood.

E. Phase 1: Single-Node Lab Demo

Use a stable illuminator and short-range controlled reflector or moving target. Record reference and surveillance channels. Demonstrate that adaptive cancellation improves the residual delay-Doppler map without erasing moving targets.

F. Phase 2: Pluto-Class Prototype

Use multiple 12-bit SDRs or a coherent multi-channel SDR with shared clocking. Prove phase coherence, time alignment, cancellation, and CAF processing. This phase is allowed to be ugly and cabled; its purpose is to expose physics and signal-processing risks.

G. Phase 3: Field Prototype

Move to a weatherproof node with reference/surveillance antennas, disciplined clock, local storage, edge compute, and network telemetry. Validate against cooperative targets, birds, vehicles, and benign drones where legally permitted.

H. Phase 4: Custom Board

Only after measured evidence supports the architecture should the project move to a custom coherent receiver board with integrated RF transceivers, FPGA/SoC, clocking, filters, and enclosure.

XIII. RISKS AND CONSTRAINTS

- **Dynamic range**: direct-path and multipath energy can clip or bury target echoes before digital processing.
- **Illuminator dependence**: transmitters may go offline, change waveform, move beams, or be unavailable where needed.

- **Geometry:** a single bistatic pair gives ambiguous ellipses; usable localization needs multiple baselines or track continuity.
 - **Synchronization:** coherent processing across channels and nodes requires disciplined clocks, calibration, and metadata.
 - **False alarms:** birds, vehicles, weather, wind turbines, and multipath can produce moving returns.
 - **Regulation:** passive listening is simpler than transmitting, but controlled transmitters require legal authorization.
 - **Export and dual-use controls:** passive bistatic or multistatic radar can fall into sensitive regulatory categories. In the United States, the eCFR text for 22 CFR Part 121 includes a category for bi-static/multi-static radar exploiting greater than 125 kHz bandwidth below 2 GHz to passively detect or track using RF transmissions such as commercial radio or television [27]. This report is not legal advice; any implementation, export, publication, or collaboration plan needs proper review.
 - **Classification data:** micro-Doppler classifiers require representative training and validation data.
- 4) **Implementation:** No hardware or software implementation was produced in this phase. The report is intended to guide simulation, lab prototypes, and later ECAD/software design.
 - 5) **Visualization:** Diagrams were added for the first-page architecture, bistatic geometry, and the candidate processing pipeline.
 - 6) **Verification:** References were selected from verified official documentation, standards pages, books, and accessible paper landing pages. URLs were tested and verification notes are included in the bibliography.

This document is a concept sketch for further technical development. It should be enriched later with measured RF data, legal/regulatory review, component quotes, simulation notebooks, field-test results, and independent engineering critique.

XIV. CONCLUSION

A cheap multistatic radar network is plausible as an engineering research program, but it is not made easy by cheap radios alone. The core technical bet is that a low-cost coherent node with enough ADC dynamic range, disciplined timing, and FPGA-based adaptive cancellation can produce useful residual signals for delay-Doppler processing and classification. The first prototype should therefore focus on measurement: how much direct-path cancellation is achievable, how many ADC bits are required, which illuminators are useful, how stable the clocks must be, and whether micro-Doppler features can separate drones from common false alarms.

ABOUT THIS DOCUMENT

This document represents a technical exploration created through a collaborative process between human and AI. The production process followed these steps:

- 1) **Discovery:** A new adjacent report workspace was requested for the topic of cheap multistatic radar.
- 2) **Initial Exploration:** The starting concept was a user-provided design prompt and a Gemini-generated engineering reality check: use broadcast, cellular, or other illuminators of opportunity; receive them with cheap coherent radio equipment; use FPGA-style streaming adaptive cancellation to suppress direct-path interference and clutter; and detect or classify weak moving echoes such as drones. The Gemini material was used as prompt/source context, not as an authoritative technical reference.
- 3) **Synthesis:** The concept was organized into a standalone systems-design document covering geometry, waveform resolution, signal models, adaptive cancellation, ADC dynamic range, compute allocation, cost modeling, and prototype phases.

NOTE ON REFERENCES AND VERIFICATION

This document contains AI-generated content. All references have been subject to verification checks to support academic and engineering integrity.

Verification Process:

- URLs were tested for accessibility.
- Author names and organizations were checked against source pages where available.
- Standards and official documentation links were preferred where possible.
- Content relevance was checked against the claims supported by each citation.

Important Notice: Readers should independently verify any references used for procurement, safety, regulatory, or operational decisions.

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